

Particulate Matter Monitoring & Health Effects

Issue 2026-08-03 | Entry window 3 August 2026 | 25 records | 9 are a dated sensing backfill — see provenance

25

RECORDS IN SCOPE

(11 EXCLUDED, 2 DUPLICATES)

10

SENSING &
INSTRUMENTATION
(40%)

5

FROM ACP / AMT
(INVISIBLE TO PUBMED)

0

EFFECT ESTIMATES
WITH A CI

6

SUBTOPIC CLUSTERS
REPRESENTED

Signal of the day

Low-cost networks have acquired the deployment longevity that multi-year exposure epidemiology requires — and, in the same window, the evidence that their drift over that horizon is uncharacterised.

Chung et al. (*Sensors*) profiled every publicly shared outdoor PurpleAir unit in the United States, 2016–2025, and find that **most remain deployed for more than three years**, densest and longest-lived in the western states. That is the first national statement that these networks are temporally continuous enough to support multi-year exposure assignment rather than campaign-scale snapshots. Read against it, Kumpika et al. (*Environ Pollut Bioavailab*) show that over exactly that horizon a PMS7003 needs a *joint* humidity-and-aging correction, not the humidity correction the literature usually stops at — and report the uncomfortable operational finding that **limiting duty cycle to 20 min/day reduces drift relative to continuous operation**, which is precisely the opposite of how a continuous exposure network is run. Silva et al. close the triangle: of 2,736 screened indoor-air studies only 24 met quality criteria, most devices fail regulatory compliance requirements, and sensor aging is named as an unaddressed confounder. The three papers were not written together, but they define the gap cleanly: *the sensors now last long enough to matter, and nobody has published the drift model that would let a five-year record be trusted end to end*. For a distributed-sensing programme that is the tractable hole in the field — not the month-one calibration problem, which is solved.

What changed today

- **Source apportionment gained an identifiability theory.** Bhardwaj & Subramanian (arXiv) stop treating sparse-network apportionment as a fitting problem and ask when the answer is *recoverable at all*: exact identifiability requires full column rank of the projected lagged response matrix, and noise robustness is governed by its singular values, coefficient visibility, background absorption, pairwise coherence and ray distance. This is the first framing in this watch that tells you whether a given sensor *placement* can distinguish two candidate sources before any data is collected — directly actionable for network siting.
- **Two independent groups did source apportionment from low-cost networks without a reference monitor.** Rehbein et al. triangulate six source clusters in Ile-Ife, Nigeria from particle-number profiles alone; Gómez et al. show in Bogotá that FastDTW alignment plus Random Forest / XGBoost with RH, temperature and absorption flow is what makes an LCS node usable against a T640X. Together these are the strongest current argument that LMIC networks need not wait for reference infrastructure.
- **The atmospheric-measurement literature reappears in force — 5 records from ACP/AMT, none of them visible to PubMed.** The Crossref by-journal leg is now carrying the part of the corpus the biomedical spine structurally cannot see, including Büttner et al.'s PIA v3.0.0: a released, versioned QA/QC pipeline for continuous INP measurement, and a template worth studying for anyone writing acceptance criteria for a sensor network.
- **Mechanistic work converged on reversibility.** Asplund et al. (*Toxicol Sci*) is the day's best-designed toxicology: concentrated ambient PM suppressed erythropoietin, reticulocytes and bone-marrow erythroid precursors and broadly depressed splenic proresolving lipid mediators (lipoxin A₄ most consistently), and *return to filtered air normalised all of it*. A recovery arm is rare in this literature and is what separates an adaptive response from damage.
- **Composition, not mass, again carries the health signal.** The Shanghai fever-clinic preprint finds cadmium the strongest single constituent, with nitrate and ammonium acting at delayed lags consistent with secondary inorganic aerosol, while Xu et al. show secondary processes dominate wintertime nitroaromatic formation across 11 Chinese cities.
- **Negative-space observation.** *No record in this window reports a ratio-metric effect estimate with a confidence interval* — Shin et al. publish no numbers in the indexed abstract — so the forest plot is omitted rather than assembled from non-comparable quantities.

Corpus analytics

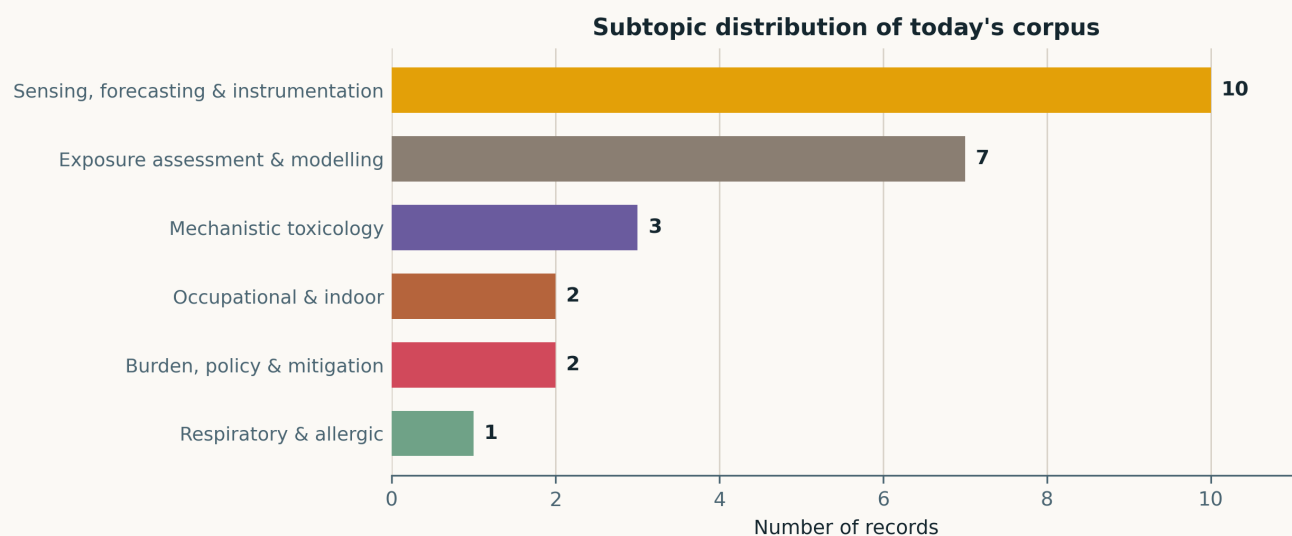


Fig. Subtopic assignment of the 25 in-scope records. The shape of this distribution is partly a sampling artefact and must not be read as a one-day flux: 9 of the 10 sensing records, and none of the health records, come from a Consensus backfill of January–June 2026 publications, run because a single Monday of PubMed entries covers almost no instrumentation. The health-side counts (3 mechanistic, 2 occupational, 2 burden, 1 respiratory) are the genuine one-day signal.

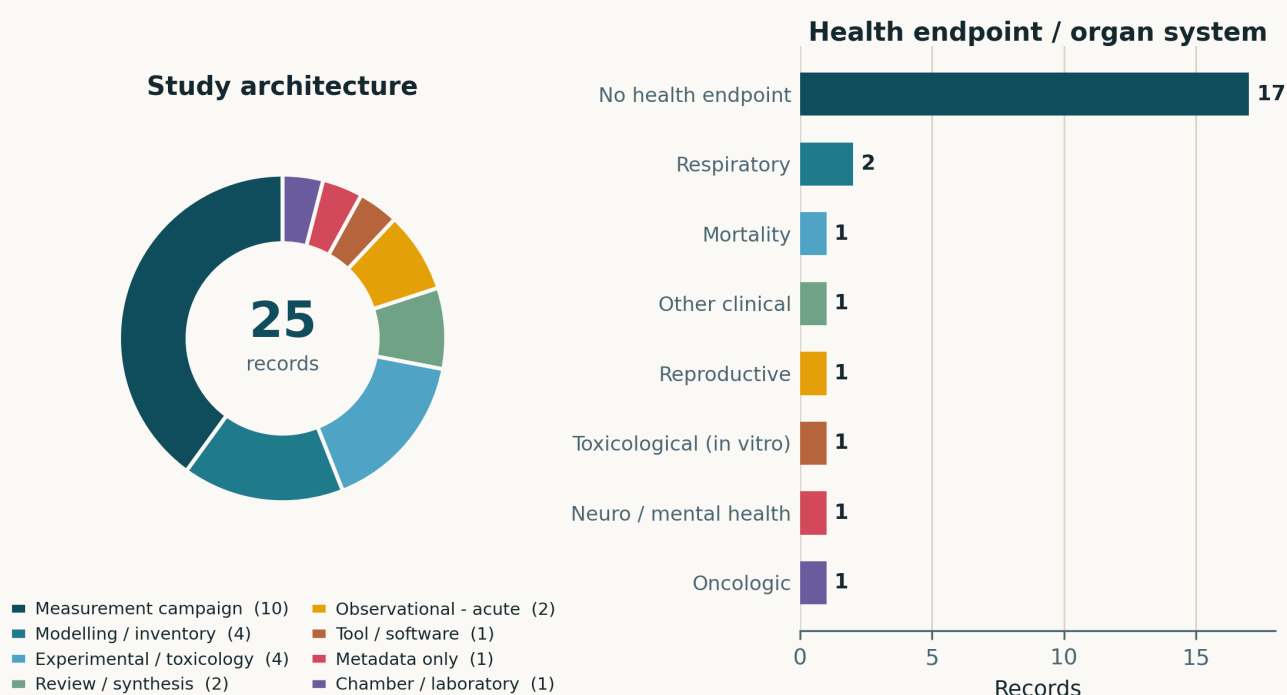


Fig. Left — study architecture. Measurement campaigns dominate at 10/25 (40%), the highest share this watch has recorded, followed by experimental toxicology and modelling/inventory at 4 each. Right — endpoint by organ system. Fourteen records carry *no* human health endpoint at all, the direct consequence of an instrumentation-weighted window; the endpoint panel should be read as thin, not as evidence of a shift in the health literature.

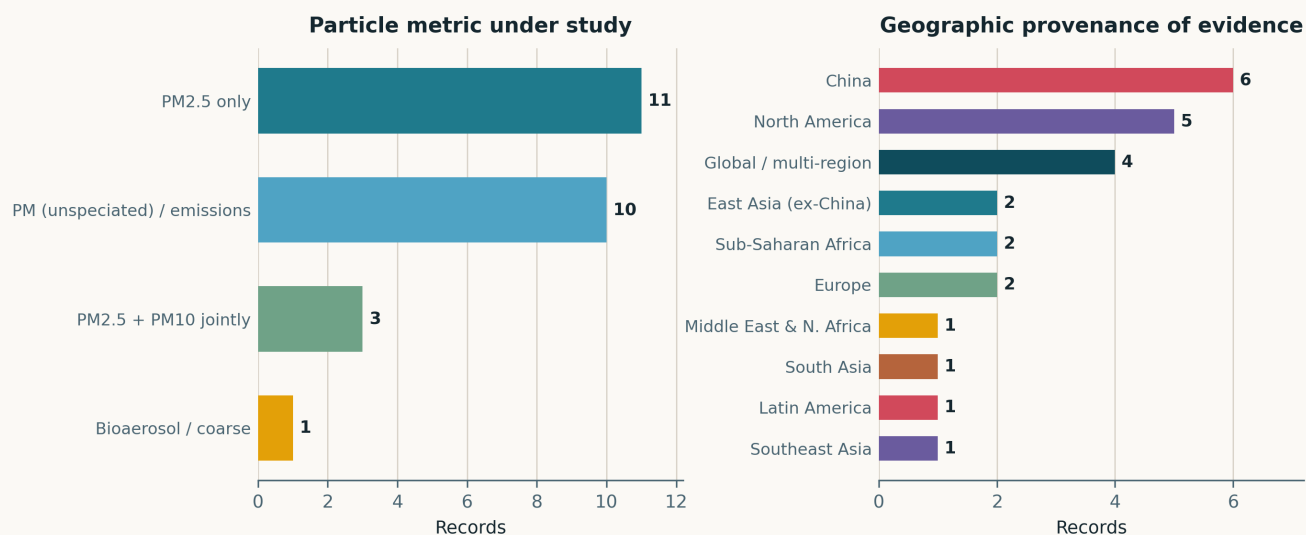


Fig. Left — particle metric. PM_{2.5} alone in 11 records; unspciated PM or emission-factor work in 10, reflecting the boiler and aerosol-chemistry records; one bioaerosol record (switchgrass pollen). Right — geographic provenance. China leads at 6 and North America at 5, but the notable feature is that **Sub-Saharan Africa appears twice** (Nigerian sensor network, Southern African dust mineralogy) with Latin America once — unusually good low- and middle-income representation for this watch, and entirely attributable to the non-PubMed legs.

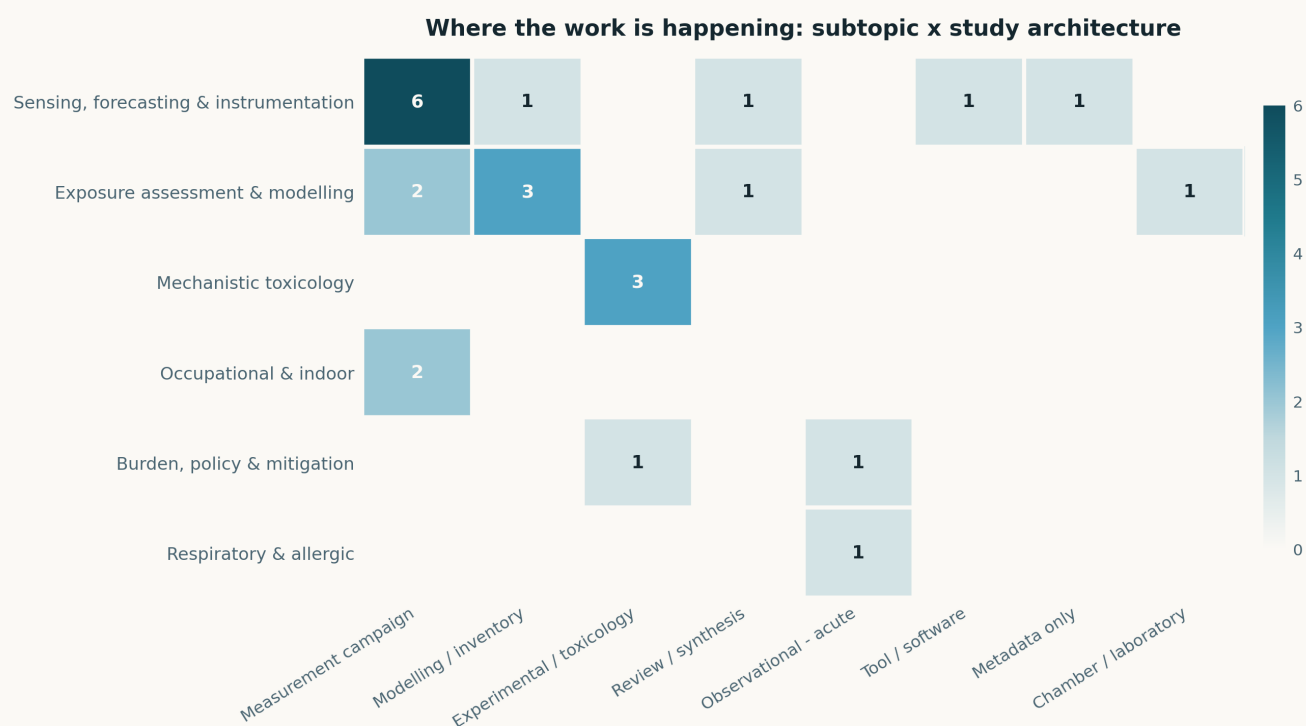


Fig. Subtopic × study-architecture cross-tabulation. The structural point is unchanged from previous issues and starker here: the *Sensing* row is entirely measurement campaigns, tool descriptions and reviews with no health endpoint, while *Mechanistic toxicology* is entirely animal work with no instrumentation content. Across 25 records there is not one study that both characterises a sensor and links it to a health outcome.

Life-course windows addressed today (records may span several stages)

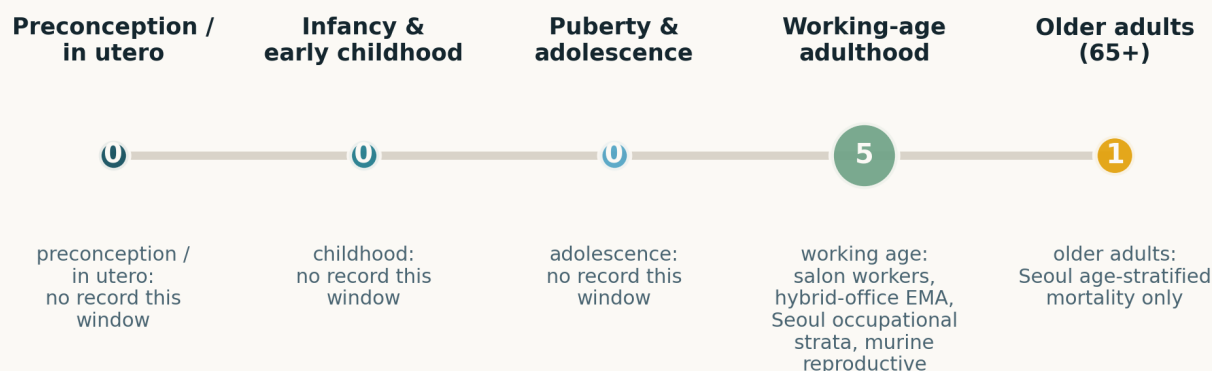


Fig. Life-course windows addressed. Counts are non-exclusive. Nineteen records are not life-course specific — instrumentation, aerosol chemistry and emission factors carry no population at all. The five working-age records are the salon workers, the hybrid-work EMA panel, the Seoul occupational strata and the murine reproductive model.

No forest plot this issue — and why that is the correct output

Not one record in this window reports a ratio-metric effect estimate with a confidence interval. Shin et al. state that blue-collar and elementary workers carry substantially higher $PM_{2.5}$ -mortality relative risks under distributed lag models, but the indexed abstract gives no point estimates; the Shanghai fever-clinic preprint reports positive constituent associations and a lag structure without published intervals. The strongest numbers in the corpus are performance statistics (Kumpika $R^2 = 0.9737$ against a BAM reference; Tsai $R^2 = 0.954$, RMSE $1.27 \mu g m^{-3}$), which are goodness-of-fit, not risk. Plotting those on a log-ratio axis would manufacture a comparison that does not exist. `EFFECTS` is empty for this date, `plots.py` therefore writes no `f5_forest.png`, and the `includegraphics` is removed from the template rather than left to compile a stale figure from a previous issue.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Sensing, forecasting & instrumentation

10 records

Chung et al. 2026 Sensors

PurpleAir deployment trends, longevity and uncertainty across the United States

Every publicly shared outdoor PurpleAir unit, 2016–2025, with deployment duration taken as the span between first and last public observation, summarised nationally, by Census region and by state. **Most sensors persist beyond three years**; density and longevity are highest in the western US. Descriptive PM_{2.5} summaries for California, Minnesota, Pennsylvania and Texas by deployment duration and urban–rural class. The limitation is that it characterises *availability*, not accuracy — persistence of a data stream is not persistence of calibration, and the paper explicitly does not address drift. That is exactly the join left open against Kumpika et al. below.

doi: 10.3390/s26061789

Kumpika et al. 2026 Environ Pollut Bioavailab

Joint humidity and sensor-aging correction for low-cost light-scattering PM_{2.5} sensors

PMS7003 units in Chiang Mai under varying RH, validated against a government BAM with additional dry-air verification through a diffusion dryer. Polynomial corrections to third order; $R^2 = 0.9737$ against BAM under elevated humidity. The operational finding is the more interesting one: limiting operation to **20 min/day materially reduces performance drift** versus continuous running. Single site, single sensor model, and the aging arm is short relative to the multi-year deployments Chung et al. document — so the correction is demonstrated rather than validated at the horizon where it matters.

doi: 10.1080/26395940.2026.2616098

Silva et al. 2026 J Air Waste Manag Assoc

Accuracy, applications and limitations of low-cost sensors for indoor air quality

PRISMA 2020 review of PubMed, Scopus and ScienceDirect to March 2025; **24 of 2,736 screened studies** met inclusion and quality criteria, appraised with the Critical Review Form for Quantitative Studies. Optical particle counters and NDIR dominate; correlations against reference instruments are strong under controlled conditions and degrade with humidity, temperature and source type. Named unaddressed confounders: sensor aging and environmental cross-sensitivity. Conclusion for the programme — most devices do not currently meet regulatory compliance requirements, and standardised validation protocols do not exist.

doi: 10.1080/10962247.2026.2624795

Gómez et al. 2026 Sensors

Machine- and deep-learning calibration of low-cost PM₁₀/PM_{2.5} nodes

LCS nodes co-located with a T640X reference at Universidad Distrital, Bogotá, with RH, temperature and absorption flow from an Automet station. Pipeline: filtering, segmentation, **FastDTW** alignment, then six calibration models (two statistical, three ML, one DL). Random Forest and XGBoost performed best; the authors argue that the FastDTW preprocessing step and the inclusion of absorption flow — not the model choice — carry the accuracy gain. Single-site co-location with no transfer test to an uncalibrated node, so generalisation is unestablished.

doi: 10.3390/s26030796

Rehbein et al. 2026 Clean Air J

Source identification and triangulation from a low-cost sensor network in Ile-Ife

Reanalysis of a published Nigerian LCS dataset using *k*-means over particle-number concentration profiles, resolving **six clusters** that separate day from night and local from regional influence. Stagnant conditions near industrial activity gave PM roughly 10× the daily mean (PM_{2.5} 30.6±40.7, PM₁₀ 83.4±259.2 µg m⁻³); cleanest conditions were daytime south-westerlies (PM_{2.5} 11.5±4.7), still above WHO annual guidance. Standard deviations exceeding the means signal a heavily skewed distribution that *k*-means handles poorly — the triangulation is indicative, not quantitative apportionment.

doi: 10.17159/caj/2026/36/1.24186

Büttner et al. 2026 Atmos Meas Tech

PIA v3.0.0: automated quality assurance for PINE ice-nucleating-particle data

Software description for the analysis chain behind the Portable Ice Nucleation Experiment. Two components documented in detail: automated data quality control, and the algorithm separating aerosols and droplets from ice crystals by optical size. The second half gives quality-assurance recommendations for PINE operation. Not a PM_{2.5} paper, but the most directly transferable artefact in the issue — a published, versioned QA/QC specification for a continuously operating particle instrument is precisely what distributed sensing networks lack.

doi: 10.5194/amt-19-5007-2026

Nimmala et al. 2026 Environ Monit Assess

Drone and ground sampling of transgenic switchgrass pollen with Lagrangian dispersal modelling

Release–recapture with 200 GE switchgrass plants expressing orange fluorescent protein under a constitutive (PvUBI1) and a pollen-specific (Zm13) promoter, sampled to **1 km** by fixed and drone-borne collectors, with Lagrangian stochastic simulation driven by high-resolution wind. Emission rate estimated by combining simulated and measured concentrations; a strong diurnal trend correlates positively with wind speed, temperature and vapour pressure deficit and negatively with RH. Incorporating changing wind direction improved estimates under low-wind meandering — the transferable result for anyone inverting a dispersion model against sparse fixed sensors. Held back from the 2 August issue on its per-record [EDAT]; it is not a gap.

doi: 10.1007/s10661-026-15632-3

Tsai et al. 2026 IEEE Sens J

Vision-based vertical PM_{2.5} retrieval from a 5G-connected UAV

Dark-channel-prior atmospheric transmittance extracted from UAV imagery, fused with meteorology into a random-forest regression with 7-D polynomial feature engineering hosted on a 5G-core virtual host; Gemini used for semantic interpretation of the vertical profile. Reported $R^2 = 0.954$, **RMSE 1.27 µg m⁻³**. Treat with caution: no held-out spatial or temporal validation is described, the DCP proxy is illumination-dependent, and the generative-AI layer contributes interpretation, not measurement. Carried for the vertical-profiling concept, not the reported accuracy.

doi: 10.1109/jsen.2026.3695609

Rubio et al. 2026 IEEE Consum Electron Mag

LoRa-connected micromobility pollution monitor for the Internet of Vehicles

On-board PM_x/CO_2 /meteorology unit for personal mobility vehicles, evaluated across Cartagena's three-gateway LoRa network. The engineering contribution is compact low-power design plus LoRa in a genuinely mobile setting, where duty-cycle limits and Doppler normally break LPWAN links. No reference co-location and no uncertainty quantification are reported, so the pollution data quality is unestablished — the communications and positioning results are the defensible part.

doi: [10.1109/mce.2026.3684542](https://doi.org/10.1109/mce.2026.3684542)

Suryawanshi et al. 2026 Int J Environ Res

Spatiotemporal variation and hotspot identification of $PM_{2.5}$ from low-cost sensors

Abstract not indexed at the time of harvest; carried as a metadata-only record so the register is complete and the next rollup does not re-surface it as new. No claims are made about its content. Full text required.

doi: [10.1007/s41742-026-01169-3](https://doi.org/10.1007/s41742-026-01169-3)

Exposure assessment & modelling

7 records

Bhardwaj & Subramanian 2026 arXiv (preprint)*Identifiability-aware source apportionment for sparse city-scale sensor networks*

Reframes inventory-constrained apportionment as a wind-conditioned lagged inverse problem: time-varying source activity is represented on a low-dimensional nonnegative temporal basis, each source-basis coefficient produces a sensor-time fingerprint, and a separate low-dimensional background space is projected out. **Exact identifiability requires full column rank of the projected lagged response matrix**; noise-robust attribution is governed by its singular values, coefficient visibility, background absorption, pairwise coherence and ray distance. The practical payoff is diagnostic — you can test before deployment whether a proposed sensor geometry can separate two inventory groups at all. Preprint, not peer reviewed.

doi: 10.48550/arXiv.2608.00050

Pavlovic et al. 2026 J Air Waste Manag Assoc*Aligning satellite Earth-observation capability, use case and capacity for air-quality health*

Twenty-three-author synthesis from the GEO Health Community of Practice Air Quality and Respiratory Health Work Group. Argues satellite EO translates into public health benefit only where three things align: technical capability of the retrieval, a defined use case, and the human, computational, operational and governance capacity to sustain a solution. Surveys applications from long-term trend tracking and epidemiological exposure assignment to short-term emissions detection and early warning, then names gaps in each dimension. A consensus document rather than new evidence — useful as a citation anchor for why a given satellite product is or is not fit for a stated purpose.

doi: 10.1080/10962247.2026.2698602

Xu et al. 2026 Atmos Chem Phys*Multi-factor control of aerosol nitroaromatic compounds across 11 Chinese cities*

Simultaneous winter measurement of PM_{2.5}-bound nitroaromatics in 11 cities. Nitrophenols and nitrocatechols dominate, with city-varying relative abundance; totals are higher in northern cities, attributed to biomass and coal combustion. Secondary processes dominate formation everywhere, but the paper's actual target is the claim — built from chamber and single-site work — that relative humidity or aerosol liquid water *drives* that formation. At large-scale field level the relationship is conditioned by multi-factor interactions rather than being a clean RH dependence. A useful corrective to chamber-derived parameterisations.

doi: 10.5194/acp-26-10751-2026

Baldo et al. 2026 Atmos Chem Phys*Chemical and mineralogical composition of Southern African mineral dust*

First comprehensive characterisation of dust from a major and under-sampled source region. Laboratory-generated aerosol from soils at natural and emerging anthropogenic source areas, characterised by Si/Al, (Ca+Mg)/Al and K/Al ratios plus clay content. Those ratios separate arid western coastal sources from humid eastern inland sources and encode source-area weathering regime. The K-feldspar finding matters beyond composition — K-feldspar is the dominant mineral ice-nucleating agent, which links this record directly to the PINE/PIA instrumentation above. Laboratory-generated, so size distribution and mineralogy need not match what is actually emitted and transported.

doi: 10.5194/acp-26-10727-2026

Wandji Nyamsi et al. 2026 Atmos Chem Phys*Broadband aerosol optical depth reconstructed from sunshine-duration records since 1900*

Uses sunshine duration as a proxy for broadband AOD together with cloud-cover observations from **2,700 sites**, producing the first historical BAOD estimates for several regions back to the late 19th century. The motivation is the persistent spread in modelled anthropogenic aerosol radiative forcing. The proxy is indirect and confounded by cloud regime and 120 years of instrument-network change; the authors' own framing — a constraint on century-scale trend, not a concentration product — is the right reading.

doi: 10.5194/acp-26-10767-2026

Wang et al. 2026 Atmos Chem Phys*Direct thermal versus emission-mediated pathways in heatwave-driven O₃ and SOA increases*

Explicitly decomposes the two routes by which heatwaves raise ozone and secondary organic aerosol: direct acceleration of photochemistry, versus temperature-driven biogenic VOC and soil-NO emissions. **Direct thermal and meteorological pathways account for over 80% (normalised)** of the national increases; ozone-temperature sensitivity peaks in the Yangtze River Delta (0.66 ppb °C⁻¹) and Pearl River Delta (0.95). Model-based attribution, so the decomposition inherits the chemical mechanism's biases — but the direction of the result argues against the common assumption that BVOC enhancement is the dominant heatwave pathway.

doi: 10.5194/acp-26-10789-2026

Li et al. 2026 Environ Monit Assess*Multi-media, multi-pathway PAH exposure and carcinogenic risk under three scenarios*

PAH concentrations in air, water, soil, dust and food compiled from Chinese studies published 2010–2024; average daily doses computed for diet, inhalation, ingestion and dermal contact; risk expressed as benzo[a]pyrene equivalents and incremental lifetime cancer risk, with Monte Carlo and sensitivity analysis. Industrial areas dominate air and water, urban areas dust and soil. **Dietary intake dominates both total exposure and carcinogenic risk**, and the industrial–urban–rural ranking is stable across dietary percentile scenarios even though absolute ILCR is not. Compiled from heterogeneous published measurements spanning 14 years — exposure misclassification is structural, not incidental.

doi: 10.1007/s10661-026-15753-9

Mechanistic toxicology

3 records

Asplund et al. 2026 Toxicol Sci*Concentrated ambient PM_{2.5} reversibly suppresses erythropoiesis and proresolving lipids*

Whole-body concentrated ambient particle exposure in mice with targeted splenic lipidomics. CAP suppressed erythropoietin, circulating reticulocytes and bone-marrow erythroid precursors, and reduced splenic iron and heme despite preserved splenic architecture — diminished erythrocyte turnover, not structural damage. Proresolving lipid mediators were broadly suppressed with **lipoxin A₄** most consistent, and its receptor ALX/FPR2 downregulated under prolonged exposure. **Return to filtered air normalised the lipid profile, LXA₄ and the erythroid parameters**. The recovery arm is what makes this the strongest design in the issue; the open question is whether the same reversibility holds after chronic rather than sub-chronic exposure.

doi: 10.1093/toxsci/kfag092

Gao et al. 2026 Zhongguo Yi Xue Ke Xue Yuan Xue Bao*SQSTM1/p62-mediated mitophagy in PM_{2.5}-aggravated asthma, and dusquetide rescue*

House-dust-mite-sensitised C57BL/6 mice exposed to PM_{2.5}, assessed by whole-body plethysmography, HE/PAS histology, TOMM20-LC3B colocalisation and LC-MS/MS mitochondrial proteomics. PM_{2.5} raised airway hyperresponsiveness, inflammation and mucus and increased mitophagy signal; Mdivi-1 inhibition reduced inflammation; proteomics identified SQSTM1/p62 as the hub, and dusquetide suppressed both p62 and mitophagy while relieving hyperresponsiveness. Mdivi-1 is a poorly selective DRP1 inhibitor with known off-target effects on complex I, so the mitophagy-specificity of the rescue is not secured by that arm alone.

doi: 10.3881/j.issn.1000-503X.16948

Deng et al. 2026 J Biochem Mol Toxicol*Baicalin rescue of PM_{2.5}-induced testicular injury via DKK1-Wnt/ β -catenin*

Oral-nasal PM_{2.5} exposure at 750 $\mu\text{g m}^{-3}$, 4 h/day for 45 days, with baicalin gavage 1 h before exposure at 50 and 100 mg/kg. PM_{2.5} lowered testicular weight and organ coefficient, sperm count and serum testosterone and downregulated StAR, CYP11A1 and CYP17A1; baicalin restored these dose-dependently alongside DKK1-Wnt/ β -catenin pathway markers. The exposure concentration sits one to two orders of magnitude above ambient even in heavily polluted settings, which is the standing weakness of this whole sub-literature — the pathway may be real, the dose is not.

doi: 10.1002/jbt.71057

Occupational & indoor**2 records****Seyedrezaei et al. 2026** J Occup Environ Med*Ecological momentary assessment plus continuous IEQ sensing across home and office*

Nineteen office workers followed four months contributing 4,559 EMAs of productivity, mood, stress and eustress–distress appraisal, against continuous CO₂, TVOC, PM_{2.5}, temperature, humidity and noise, analysed with mixed-effects and multilevel mediation. Office work associated with higher productivity and greater eustress–distress coexistence, home with higher boredom and modestly lower stress; **home workspaces had poorer air quality**, and CO₂ and TVOC indirectly explained part of the mood and stress difference. $n = 19$ is very small for multilevel mediation, and the within-person design cannot separate the environment from everything else that differs between home and office — read the IEQ mediation as hypothesis-generating. The methodological interest is the pairing of dense subjective sampling with dense environmental sampling, which is the design template for sensor-linked wellbeing work.

doi: 10.1097/JOM.0000000000003833

Niari et al. 2026 J Environ Health Sci Eng*Microplastic burden in beauty-salon indoor air and worker nasal lavage fluid*

Thirty-six indoor air and 18 nasal lavage samples across salons and homes in Ardabil. Salon air carried 99.4 $\pm$ 4.1 versus 36.6 $\pm$ 3.1 MP m⁻³ at home, with nasal lavage 12.3–23.8 versus 9.3–20.1 MP mL⁻¹ and estimated daily exposure 572 $\pm$ 238 versus 212 $\pm$ 109 MP/d. The nasal-lavage arm is the useful part — an internal-dose measurement rather than an inferred one — but the sample is small, no procedural blank is reported for a matrix acutely prone to airborne plastic contamination, and no polymer identification is described. Directionally credible, quantitatively soft.

doi: 10.1007/s40201-026-00999-9

Respiratory & allergic**1 record****Wang et al. 2026** Res Sq (preprint)*PM_{2.5} constituents and daily fever-clinic visits in Jinshan, Shanghai*

Time-series generalized additive models linking 77,637 fever-clinic visits in 2024 to PM_{2.5} mass and speciated constituents, with single-day lags 0–7 and cumulative moving averages 01–07, adjusted for seasonality, day of week, holidays and meteorology, stratified by sex, age and season. Mean PM_{2.5} 32.0 $\mu\text{g m}^{-3}$. PM_{2.5}, arsenic, cadmium, chromium, lead, manganese, nitrate and ammonium were positively associated; **cadmium showed the largest association** while nitrate and ammonium acted at delayed lags consistent with secondary inorganic aerosol. Preprint, not peer reviewed; no effect sizes or intervals in the abstract; constituents are highly collinear and single-pollutant time-series models will attribute shared variance arbitrarily among them. Fever-clinic attendance is also a behavioural endpoint sensitive to concurrent epidemic activity.

doi: 10.21203/rs.3.rs-10211219/v1

Burden, policy & mitigation**2 records****Galvin et al. 2026** J Air Waste Manag Assoc*Cyclic-operation emission factors for biodiesel, renewable diesel and hydrogen-blend boilers*

Emission factors developed under *representative cyclic operation* for three appliances — a residential retention-head liquid-fuel boiler, a residential modulating-condensing gas boiler and a two-stage commercial boiler — across B20, B100, renewable diesel, natural gas and a 20% hydrogen blend. PM measured by EPA Methods 5 and 202 to capture filterable and condensable fractions, supplemented with real-time instrumentation. Two results matter: **burner on-off cycling drives incomplete-combustion pollutants and is not represented in current inventory factors**, which derive from steady-state operation of larger plant; and **condensable PM can dominate total PM**, especially in advanced gas systems. An optical sensor sited near residential combustion is blind to the condensable fraction at the stack and sees it only once it forms — a real limitation for source attribution near buildings.

doi: 10.1080/10962247.2026.2657442

Shin et al. 2026 J Occup Environ Med*Occupational disparities in PM_{2.5}-related mortality under distributed lag models*

Five years of daily all-cause mortality and PM_{2.5} for Seoul, analysed with generalized additive plus distributed lag models and stratified by age, sex and occupation. A significant positive PM_{2.5}–mortality association overall, with **blue-collar and elementary workers carrying substantially higher relative risks**. The policy argument — that concentration targets alone are insufficient and high-vulnerability strata need tailored management — is well posed, but the indexed abstract reports *no point estimates or intervals*, so the magnitude cannot be assessed and the record contributes no effect estimate. Occupation comes from death certificates, which misclassifies the late-career and retired.

doi: 10.1097/JOM.0000000000003831

Corpus register

First author	Focus	Journal	Design	Metric	Region
Respiratory & allergic					
Wang et al.	PM _{2.5} constituents and daily fever-clinic visits: a time-series study in Jinshan, Shanghai	<i>Res Sq (preprint)</i>	Time-series	PM _{2.5} mass + As, Cd, Cr, Pb, Mn, nitrate, ammonium	Shanghai, China
Mechanistic toxicology					
Asplund et al.	Concentrated ambient PM _{2.5} suppresses erythropoiesis and splenic proresolving lipid mediators, reversibly	<i>Toxicol Sci</i>	Murine inhalation exposure	Concentrated ambient particles (whole-body)	United States (Louisville CAP facility)
Deng et al.	Baicalin rescue of PM _{2.5} -induced testicular injury via DKK1-Wnt/beta-catenin signalling	<i>J Biochem Mol Toxicol</i>	Murine inhalation exposure	PM _{2.5} , 750 µg/m ³ , 4 h/day, 45 days (oral-nasal)	China
Gao et al.	SQSTM1/p62-mediated mitophagy in PM _{2.5} -aggravated HDM asthma, and duscetide rescue	<i>Zhongguo Yi Xue Ke Xue Yuan Xue Bao</i>	Murine inhalation exposure	PM _{2.5} (HDM-sensitised C57BL/6)	China
Occupational & indoor					
Niari et al.	Microplastic burden in beauty-salon indoor air and worker nasal lavage fluid	<i>J Environ Health Sci Eng</i>	Observational exposure study	Airborne microplastics (MPs/m ³); nasal lavage MPs/mL	Ardabil, Iran
Seyedrezaei et al.	Ecological momentary assessment and continuous IEQ sensing across home and office workspaces	<i>J Occup Environ Med</i>	Repeated-measures biomarker	PM _{2.5} , CO ₂ , TVOC, T, RH, noise	United States
Sensing, forecasting & instrumentation					
Buettner et al.	PIA v3.0.0: automated quality assurance and analysis for PINE ice-nucleating-particle data	<i>Atmos Meas Tech</i>	Software / tool description	INP concentration; optical particle size discrimination	Germany (multi-site PINE)
Chung et al.	PurpleAir deployment trends, longevity and uncertainty across the United States, 2016-2025	<i>Sensors</i>	Network deployment + field evaluation	PM _{2.5} (PurpleAir PA-II)	United States (national)
Gomez et al.	Machine- and deep-learning calibration of low-cost PM ₁₀ /PM _{2.5} nodes against a T640X reference	<i>Sensors</i>	Calibration model comparison	PM ₁₀ , PM _{2.5} (LCS vs T640X)	Bogota, Colombia
Kumpika et al.	Joint humidity and sensor-aging correction for low-cost light-scattering PM _{2.5} sensors	<i>Environ Pollut Bioavailab</i>	Mechanistic correction framework	PM _{2.5} (PMS7003 vs BAM reference)	Chiang Mai, Thailand
Nimmala et al.	Drone and ground sampling of transgenic switchgrass pollen with Lagrangian dispersal modelling	<i>Environ Monit Assess</i>	Release-recapture field experiment	Bioaerosol (OFF-tagged pollen), high/low-volume samplers	Oliver Springs, TN, USA
Rehbein et al.	Source identification and triangulation from a low-cost sensor network in Ile-Ife	<i>Clean Air J</i>	Source apportionment	PM _{2.5} , PM ₁₀ , particle number concentration	Ile-Ife, Nigeria
Rubio et al.	LoRa-connected micromobility pollution monitor for the Internet of Vehicles	<i>IEEE Consum Electron Mag</i>	Proof-of-concept deployment	PM _x , CO ₂ , meteorology	Cartagena, Spain
Silva et al.	Accuracy, applications and limitations of low-cost sensors for indoor air quality	<i>J Air Waste Manag Assoc</i>	Systematic review (PRISMA)	PM _{2.5} , PM ₁₀ , CO ₂ (OPC and NDIR)	Global
Suryawanshi et al.	Spatiotemporal variation and hotspot identification of PM _{2.5} from low-cost sensors	<i>Int J Environ Res</i>	Metadata only (no abstract)	PM _{2.5} (low-cost sensors)	India (urban)
Tsai et al.	Vision-based vertical PM _{2.5} retrieval from a 5G-connected UAV with generative-AI interpretation	<i>IEEE Sens J</i>	Proof-of-concept deployment	PM _{2.5} (dark-channel-prior transmittance proxy)	Taiwan
Exposure assessment & modelling					
Baldo et al.	Chemical and mineralogical composition of Southern African mineral dust	<i>Atmos Chem Phys</i>	Laboratory metrology	Mineral dust (Si/Al, (Ca+Mg)/Al, K/Al, clay)	Southern Africa
Bhardwaj & Subramanian	Identifiability-aware source apportionment for sparse city-scale sensor networks	<i>arXiv (preprint)</i>	Source apportionment	PM (wind-conditioned, lagged inverse problem)	Method (city-scale, synthetic + real)
Li et al.	Multi-media multi-pathway PAH exposure and carcinogenic risk under industrial, urban and rural scenarios	<i>Environ Monit Assess</i>	Physical exposure model	PAHs in air, water, soil, dust, food (BaP-eq)	China
Pavlovic et al.	Aligning satellite Earth-observation capability, use case and capacity for air-quality health decisions	<i>J Air Waste Manag Assoc</i>	Mapping review	Satellite AOD-derived PM _{2.5} , trace gases	Global (GEO Health CoP)

First author	Focus	Journal	Design	Metric	Region
Wandji Nyamsi et al.	Broadband AOD reconstructed from sunshine-duration records since 1900	<i>Atmos Chem Phys</i>	Physical exposure model	Broadband aerosol optical depth	Global (2,700 sites)
Wang et al.	Direct thermal vs emission-mediated pathways in heatwave-driven O ₃ and SOA increases	<i>Atmos Chem Phys</i>	Chemical transport model	SOA, O ₃ (BVOC and soil-NO pathways)	China
Xu et al.	Multi-factor control of aerosol nitroaromatic compounds across 11 Chinese cities in winter	<i>Atmos Chem Phys</i>	Supersite observation	PM _{2.5} -bound nitrophenols and nitrocatechols	11 cities, China
Burden, policy & mitigation					
Galvin et al.	Cyclic-operation emission factors for biodiesel, renewable diesel and hydrogen-blend boilers	<i>J Air Waste Manag Assoc</i>	Chamber emission characterisation	Filterable + condensable PM (EPA M5/M202) + real-time PM	United States
Shin et al.	Occupational disparities in PM _{2.5} -related mortality under distributed lag models	<i>J Occup Environ Med</i>	Time-series	PM _{2.5} (daily, city-wide)	Seoul, South Korea

Provenance and method

Entry window 3 August 2026, contiguous with the 2 August issue. PubMed (connector plus the local E-utilities harvester) was queried on two separate axes against [Date - Entry]: health (11 hits) and instrumentation (6 hits, 5 of them already in the health set), union 12 after deduplication, of which **10 are carried**. Nimmala et al. was held back from the 2 August issue on its per-record [EDAT] and is picked up here as planned — it is not a gap. **Europe PMC** returned 5, contributing one preprint PubMed does not index. **Crossref by-journal** returned 5 from *Atmos. Chem. Phys.* and *Atmos. Meas. Tech.*, all carried; PubMed indexes neither journal, which is the entire reason that leg exists. **arXiv** yielded 1 record inside the recency screen (1 further hit rejected as off-topic). **OpenAlex** returned **HTTP 429** on top of its standing paid-plan block — the leg remains dead and that coverage is routed through Consensus.

The 9 records marked as backfill are not 3 August entries. They are January–June 2026 publications surfaced by four Consensus sweeps (multi-year drift; network source apportionment; mobile and hyperlocal monitoring; personal exposure) plus one arXiv preprint of 25 July, run because a single Monday of PubMed entries covers essentially no instrumentation. They are dated to this issue for bookkeeping and flagged here, in the masthead and in the f1 caption. **The next weekly rollup must not read a trend through them.** Consensus returned at most 3 results per sweep on this run (unauthenticated free tier), which is the binding constraint on the sensing leg and is worth resolving before the next backfill.

Excluded: 11 records logged with reasons to `state/rejected.jsonl` — 2 duplicates already carried on 2 August (Jahanshahi, dementia onset; Thies, London Underground commuting), 3 off-scope catalysis or bibliometrics papers, 3 out-of-recency Consensus returns (2018–2021), 1 off-topic arXiv statistics paper, 1 excluded on sensor-quality grounds, and **1 in-scope record dropped for lack of a resolvable DOI** — the Canadian wildfire PM_{2.5} spatiotemporal analysis (PMID 42544279) carries a PMC identifier only, the NCBI ID converter returns no DOI and Crossref does not know the title, so no link could be shipped.

All 25 shipped DOIs passed `check_dois.py` (0 fail, 8 advisory paraphrase or not-in-Crossref warnings). First-author surnames are read from the `efetch AuthorList` or the Crossref record, never inferred. Subtopic, design, metric and geography labels are single-label editorial judgements, not a controlled vocabulary. Content derived from publisher metadata; abstracts remain the copyright of their respective publishers.